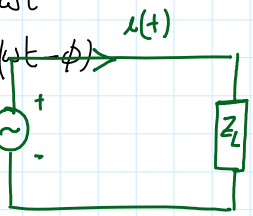
Lecture 15: Single Phase AC Circuits (Contd...)

Recapitulation: 1φ system: $v(t) = V_m \sin \omega t$

$i(t) = I_m \sin(\omega t - \phi)$



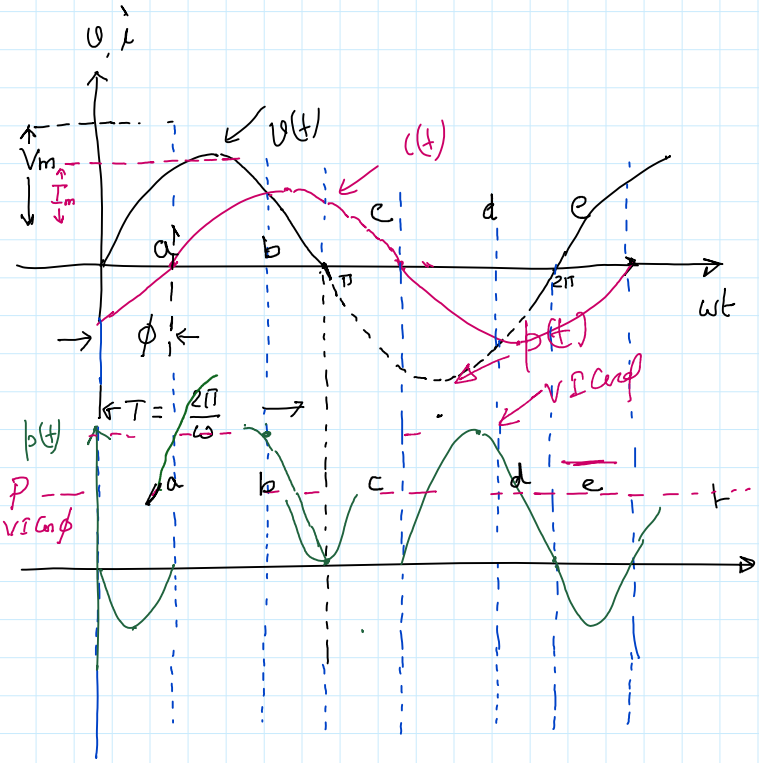
Average value: $V_{avg} = \frac{1}{\pi}\int_0^{\pi} v(t)\,dt = \frac{2V_m}{\pi}$

$I_{avg} = \frac{1}{\pi}\int_0^{\pi} i(t)\,dt = \frac{2I_m}{\pi}$

RMS = Root Mean Square value:

$$V_{rms} = \left[\frac{1}{T}\int_0^T v^2(t)\,dt\right]^{1/2} = \frac{V_m}{\sqrt{2}}$$

$$I_{rms} = \left[\frac{1}{T}\int_0^T i^2(t)\,dt\right]^{1/2} = \frac{I_m}{\sqrt{2}}$$



Powers: $p(t) = v(t)\,i(t)$

$$= \frac{V_m I_m}{2} \cdot 2\sin\omega t \sin(\omega t - \phi)$$

$$= \frac{V_m I_m}{2}\left[\cos\phi - \cos(2\omega t - \phi)\right]$$

$$p(t) = \frac{V_m I_m}{2}\cos\phi\,(1 - \cos 2\omega t) - \frac{V_m I_m}{2}\sin\phi \sin 2\omega t$$

$$= p_{unidirectional} + p_{oscillating}$$

$$= p_{active}(t) + p_{reactive}$$

Real Power, $P = \frac{1}{T}\int_0^T p(t)\,dt = \frac{1}{T}\int_0^T p_{active}(t) = \frac{V_m I_m}{2}\cos\phi$

$$\boxed{P = VI\cos\phi}$$

where $V_m/\sqrt{2} = V_{rms} = V$

$I_m/\sqrt{2} = I_{rms} = I$

Reactive Power $Q \triangleq$ Maximum value of $p_{oscillation} = \frac{V_m I_m}{2}\sin\phi$

$$\boxed{Q = VI\sin\phi}$$

Apparent Power (S): Product of RMS values of voltage and current.

$$\boxed{S = V_{rms} I_{rms} = VI} = \sqrt{P^2 + Q^2}$$

Power Factor (pf): Ratio of real power to the apparent power.

$$pf = \frac{P}{S} = \frac{VI\cos\phi}{VI} = \cos\phi$$

$$pf = \frac{P}{S} = \frac{VI\cos\phi}{VI} = \cos\phi$$

$$\boxed{pf = \cos\phi}$$

ϕ = angle between the current and the voltage.

Relationship between P, Q, S, pf is illustrated through Power Triangle.

$$S = VI = \sqrt{(VI\cos\phi)^2 + (VI\sin\phi)^2}$$

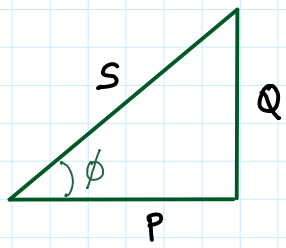
$$\boxed{S = \sqrt{P^2 + Q^2}}$$

or $S^2 = P^2 + Q^2$

From right angled triangle,

$$\cos\phi = \frac{P}{S}, \quad \sin\phi = \frac{Q}{S}$$

$$\tan\phi = \frac{Q}{P}$$



Power triangle

Phasor Representation of AC quantities:

$$i(t) = I_m \sin\omega t = \sqrt{2} I_{rms} \sin\omega t$$

$$i(t) = \sqrt{2} I \sin\omega t = \text{Im}\left(I_m e^{j\omega t}\right)$$

$$= I_m \left[\text{Im}(e^{j\omega t})\right]$$

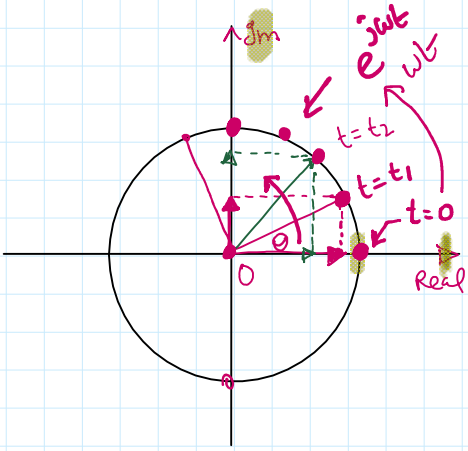
(Im: imaginary; I_m: peak value)

Euler's formula:

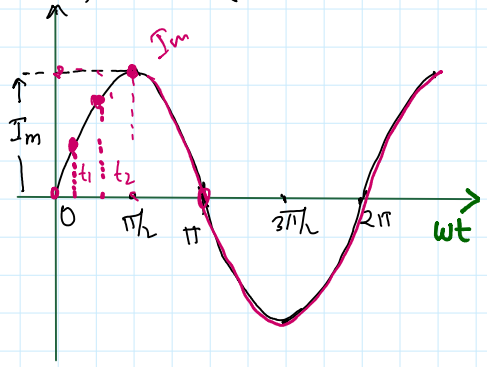
$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$e^{j\omega t} = \cos\omega t + j\sin\omega t$$

How do we understand the term: $e^{j\omega t}$, especially $e^{j\omega t} = 1\angle\omega t$



$$i(t) = I_m \text{Im}(e^{j\omega t}) = \sqrt{2} I \text{Im}(e^{j\omega t})$$



$i(t) = I_m \sin\omega t = \sqrt{2} I \sin\omega t$ (time domain)

Thus in phasor form, we write: $\bar{I} = I\angle 0^\circ$ A ← (phasor)

Thus in phasor form, we write: $\bar{I} = I \angle 0^\circ$ A ← (phasor)

where $I = |\bar{I}|$ = Rms value of current

$\phi = \angle \bar{I} = 0$ = Angle of $\bar{I}$ with respect to the reference phasor

In general, $i(t) = I_m \sin(\omega t + \phi)$

$$= \text{Im}\left[I_m e^{j(\omega t+\phi)}\right] = \text{Im}\left\{I_m\left(e^{j\omega t} \cdot e^{j\phi}\right)\right\}$$

$$= \text{Im}\left(I_m e^{j\phi} \cdot e^{j\omega t}\right)$$

$$= \text{Im}\left[\left(\bar{I}_m\right) \cdot e^{j\omega t}\right]$$

$$= \text{Im}\left(I_m \angle \phi \cdot e^{j\omega t}\right)$$

Circuit

$i(t) = \sqrt{2}\, I \sin(\omega t + 90^\circ)$

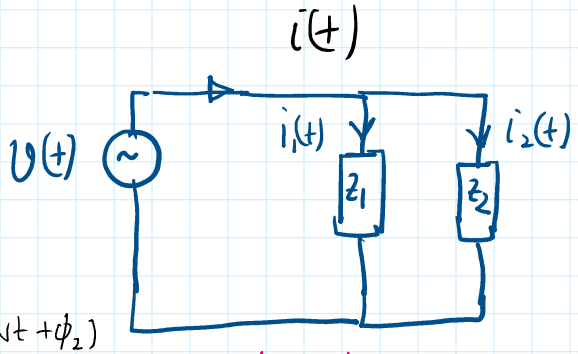
$\bar{I} = I \angle 90^\circ$

$$\bar{I}_m = I_m e^{j\phi} = I_m \angle \phi$$

$$\frac{\bar{I}_m}{\sqrt{2}} = \frac{I_m}{\sqrt{2}} e^{j\phi} = \frac{I_m}{\sqrt{2}} \angle \phi$$

$$\bar{I} = I e^{j\phi} = I \angle \phi$$

Example: Consider the following circuit. Find out the total current $i(t)$!



For $v(t) = V_m \sin \omega t$ (Reference phasor)

$i_1(t) = I_{m1} \sin(\omega t + \phi_1)$

$i_2(t) = I_{m2} \sin(\omega t + \phi_2)$

The total current $i(t)$ is given by

$$i(t) = i_1 + i_2 = I_{m1} \sin(\omega t + \phi_1) + I_{m2} \sin(\omega t + \phi_2)$$

$$= I_{m3} \sin(\omega t + \phi_3)$$

I_{m3} ?, ϕ_3 = ?

$$= I_{m1} \sin\omega t \cos\phi_1 + I_{m1} \cos\omega t \sin\phi_1 + I_{m2} \sin\omega t \cos\phi_2 + I_{m2} \cos\omega t \sin\phi_2$$

$$= (I_{m1} \cos\phi_1 + I_{m2} \cos\phi_2) \sin\omega t + (I_{m1} \sin\phi_1 + I_{m2} \sin\phi_2) \cos\omega t$$

$$= I_{m3} \cos\phi_3 \sin\omega t + I_{m3} \sin\phi_3 \cos\omega t$$

$$= I_{m3}\cos\phi_3 \sin\omega t + I_{m3}\sin\phi_3 \cos\omega t$$

$$i(t) = I_{m3}\sin(\omega t + \phi_3)$$

Here $I_{m3}\cos\phi_3 = I_{m1}\cos\phi_1 + I_{m2}\cos\phi_2$ ---- (1)

$I_{m3}\sin\phi_3 = I_{m1}\sin\phi_1 + I_{m2}\sin\phi_2$ ----- (2)

Squaring and adding gives I_{m3}, dividing (2) by (1) gives ϕ_3.

Where $$I_{m3} = \left[\left(I_{m1}\cos\phi_1 + I_{m2}\cos\phi_2\right)^2 + \left(I_{m1}\sin\phi_1 + I_{m2}\sin\phi_2\right)^2\right]^{\frac{1}{2}}$$

$$\phi_3 = \tan^{-1}\frac{I_{m1}\sin\phi_1 + I_{m2}\sin\phi_2}{I_{m1}\cos\phi_1 + I_{m2}\cos\phi_2}$$

This needs considerable computation. The alternate way is to use phasor, as explained previously. For example

$$i(t) = i_1 + i_2$$
$$= \mathrm{Im}\left\{I_{m1}e^{j(\omega t+\phi_1)}\right\} + \mathrm{Im}\left\{I_{m2}e^{j(\omega t+\phi_2)}\right\}$$
$$= \mathrm{Im}\left[I_{m1}e^{j(\omega t+\phi_1)} + I_{m2}e^{j(\omega t+\phi_2)}\right]$$
$$= \mathrm{Im}\left[I_{m1}e^{j\phi_1}e^{j\omega t} + I_{m2}e^{j\phi_2}e^{j\omega t}\right]$$
$$= \mathrm{Im}\left[\left(I_{m1}e^{j\phi_1} + I_{m2}e^{j\phi_2}\right)e^{j\omega t}\right]$$
$$= \mathrm{Im}\left[\left(\bar{I}_{m1} + \bar{I}_{m2}\right)e^{j\omega t}\right]$$
$$= \mathrm{Im}\left[\bar{I}_{m3}e^{j\omega t}\right]$$

Where $\bar{I}_{m3} = \bar{I}_{m1} + \bar{I}_{m2}$

$$\frac{\bar{I}_{m3}}{\sqrt{2}} = \frac{\bar{I}_{m1}}{\sqrt{2}} + \frac{\bar{I}_{m2}}{\sqrt{2}}$$

$$\boxed{\bar{I}_3 = \bar{I}_1 + \bar{I}_2}$$

This is known as phasor sum of two currents.

Once phasor is computed, we can find time domain expression as follows:

$$i_3(t) = \sqrt{2}\,|\bar{I}_3|\sin(\omega t + \angle\bar{I}_3)$$

$$i_3(t) = \sqrt{2}\, I_3 \sin(\omega t + \phi_3)$$

Example 1: In above circuit, consider,

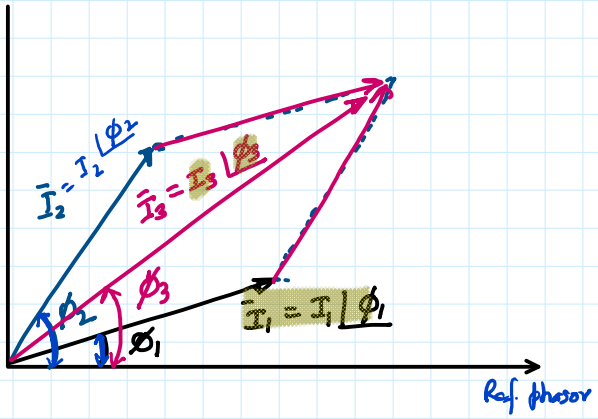
$i_1 = 10\sqrt{2}\sin\omega t$

$i_2 = 20\sqrt{2}\sin(\omega t + 60^\circ)$

Find current $i_3 = i_1 + i_2$

$i_1 = 10\sqrt{2}\cos\omega t$

$i_2 = 20\sqrt{2}\cos(\omega t + 60^\circ)$



$$\bar{I}_1 = 10 \angle 0$$
$$\bar{I}_2 = 20 \angle 60^\circ$$

$$\bar{I}_3 = \bar{I}_1 + \bar{I}_2 = 10\angle 0 + 20\angle 60^\circ$$
$$= 10 + j0 + 20 \times \tfrac{1}{2} + j20\,\tfrac{\sqrt{3}}{2}$$
$$= 20 + j\,10\sqrt{3} = 20 + j\,17.32$$
$$= \sqrt{20^2 + 17.32^2} \angle \tan^{-1}\tfrac{17.32}{20}$$
$$\bar{I}_3 = 26.45 \angle 41^\circ \ A$$
$$= I_3 \angle \phi_3$$
$$i_3(t) = \sqrt{2} \times 26.45 \sin(\omega t + 41^\circ)$$

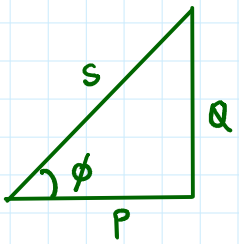
Power Expression Using Voltage and Current Phasors:

we define a term called complex apparent power, $\bar{S}$ as

$$\bar{S} = \bar{V}\,\bar{I}^*$$
$$= V\angle 0 \,(I\angle\phi)^*$$
$$= VI\angle -\phi$$
$$= VI\cos\phi + j(-VI\sin\phi)$$
$$\bar{S} = P + jQ$$

$$S = |\bar{S}| = \sqrt{P^2 + Q^2}$$

For example: $\bar{V} = V\angle 0$, $\bar{I} = I\angle\phi$



Power triangle

Example 2: Consider the following voltage and current

$$v(t) = 170\sin(\omega t + 10^\circ)$$

$i(t) = \ldots$ for $\omega = 314$ rad/sec for ...

$v(t) = 170 \sin(\omega t + 10^\circ)$

$i(t) = 14.14 \sin(\omega t - 20^\circ)$ for $\omega = 314$ rad/sec for $f = 50$ Hz

$20 - (-10^\circ)$
$\phi = 30^\circ$

a) Represent voltage and current in phasor form

b) Determine the average power (real power), reactive power, apparent power, power factor.

c) Express complex apparent power $\bar{S}$.

Solution:

$$\bar{V} = \frac{170}{\sqrt{2}} \angle 10^\circ$$

$$\bar{I} = \frac{14.14}{\sqrt{2}} \angle -20^\circ$$

$$\bar{S} = \bar{V}\bar{I}^* = \frac{170 \times 14.14}{2} \angle 30^\circ = 1201.9 \angle 30^\circ$$

$$= 1040.87 + j\,600.95$$

$\Rightarrow P = 1040.87$ W $= P_{avg}$

$Q = 600$ VAr